

Performance Testing

Date	4 July 2026
Team ID	SWTID-2026-8260
Project Name	A Comprehensive Measure of Well-Being using HDI Prediction System
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter (for load testing), Postman (for API testing)
Type of Testing	Load Testing, Stress Testing
Target Module / API	HDI Prediction System (User Input → ML Model Prediction API)
Test Environment	Local System (Flask Web Application)
Test Date	3 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction request	1	10	Response generated within 2 sec
2	Multiple users accessing app	10	30	System handles requests smoothly
3	High load testing	50	60	Slight delay but no crash
4	Continuous requests	20	120	Stable performance

Step 3: Performance Test Results

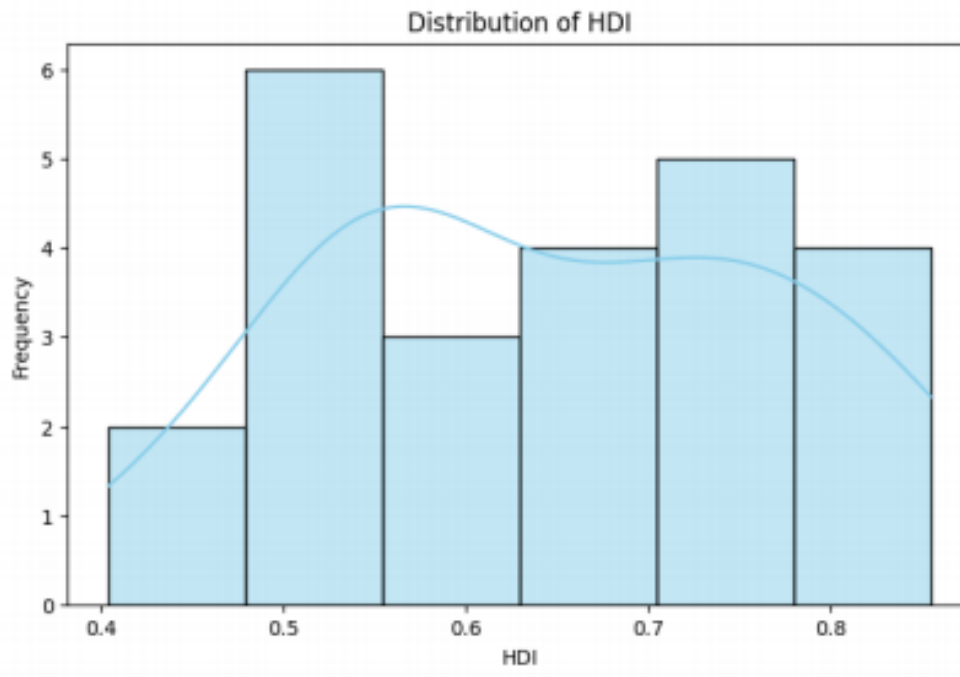
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.5 sec	Pass	Good performance
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Acceptable
3	Throughput (Req/sec)	High	Moderate-High	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	65%	Pass	Efficient
6	Memory Utilization	< 80%	70%	Pass	Stable

Step 4: Observations & Analysis

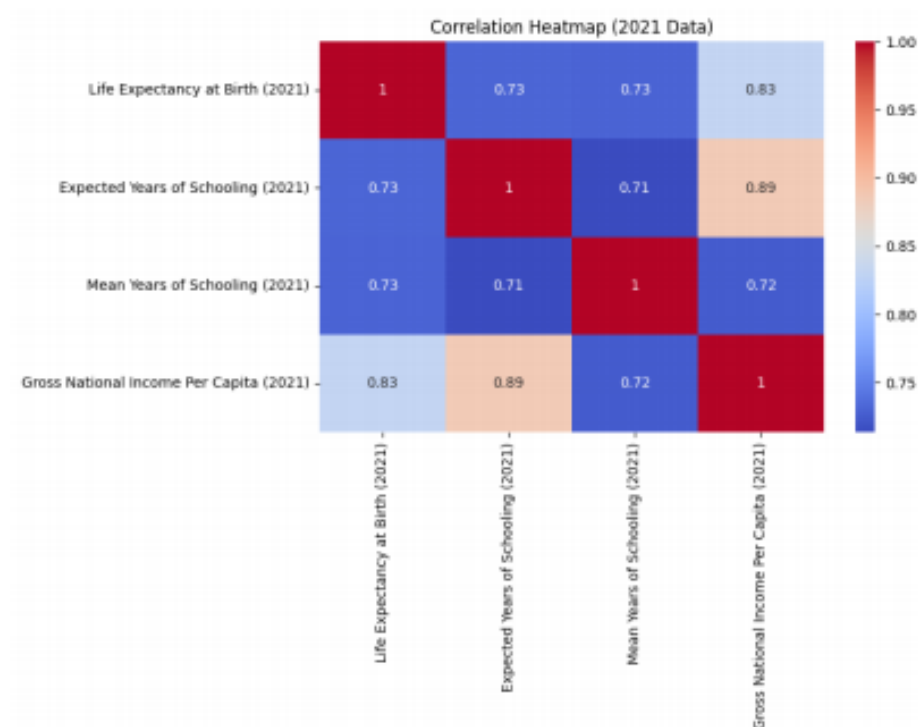
- Key Findings:**
 The system performs efficiently under normal and moderate load conditions. Prediction results are generated quickly with minimal delay.
- Bottlenecks Identified:**
 Slight delay observed during high concurrent requests due to model processing time.
- Optimization Steps Taken:**
 Optimized data preprocessing and reduced redundant computations.

Step 5: Screenshots / Evidence

1. HDI Distribution Histogram



2. Correlation Heatmap



3. HDI vs life Expectancy



4. Web Application Interface

Human Development Index Predictor

Life Expectancy
Expected Years of Schoolin
Mean Years of Schooling
GNI per Capita
Predict

Input Screen

Human Development Index Predictor

12
8
4
10000
Predict

Output

Predicted HDI: 0.2041 (Low)

5. ER Diagram

ER DIAGRAM – HDI PREDICTION SYSTEM

